

WEEK 8

Student Performance Dashboard

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BSC DATA SCIENCE & ARTIFICIAL INTELLIGENCE

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1. Introduction and Objectives

This report documents the Week 8 task, in which the data model prepared in Week 7 was turned into an interactive Student Performance Dashboard in Power BI Desktop. The dashboard allows a reader to move from a high-level summary of academic performance to the detail of individual student profiles, and to explore how background factors relate to scores.

1.1 Aim

To design, build, format, validate and interpret a dashboard that answers key academic questions about student performance using the Math, Reading and Writing scores in the dataset.

1.2 Objectives

- Verify the Week 7 data model and DAX measures before building visuals.
- Create KPI cards for the average score, subject averages and overall pass percentage.
- Build visuals comparing subjects, parental education, test preparation, lunch type and grade distribution.
- Add slicers so that the reader can filter by gender, race/ethnicity, parental education, lunch type, test preparation and subject.
- Create a drill-through page named Academic Profile with a Back button.

2.Dataset and Data Model Verification

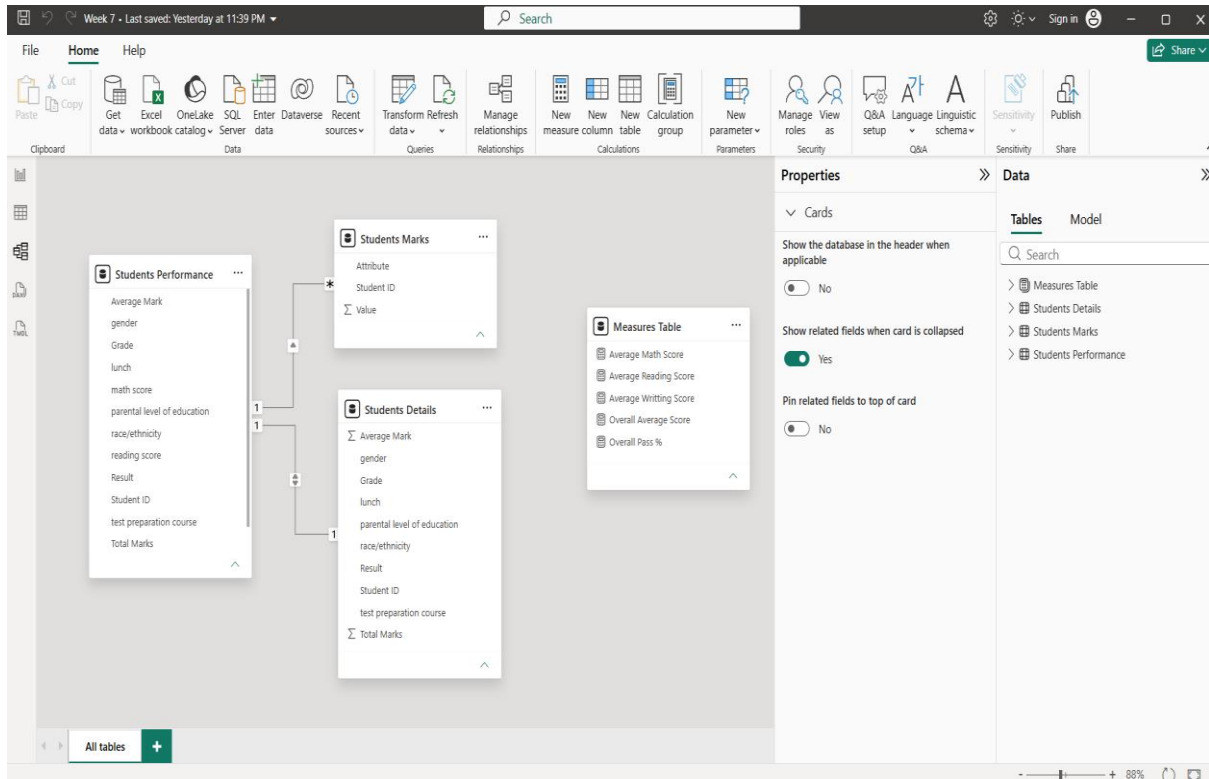
2.1 Tables in the Model

Table	Key fields / measures
Students Details	Student ID, gender, race/ethnicity, parental level of education, lunch, test preparation course, Grade, Result, Total Marks, Average Mark
Students Marks	Student ID, Attribute (math score, reading score, writing score) and Value, the unpivoted form used for subject comparisons
Students Performance	Average Mark, gender, Grade, lunch, math score, parental level of education, race/ethnicity, reading score, Result, test preparation course, Total Marks, writing score
Measures	Average Math Score, Average Reading Score, Average Writing Score, Overall Average Score, Overall Pass %

2.2 Verification Checks

- Confirmed that the score columns (math, reading, writing) are whole numbers with Summarization set to Don't summarize, so that averages come from explicit measures .
- Confirmed that Student ID links the tables and that relationships are present in Manage relationships.

- Confirmed that the measures return plausible values (subject averages between 0 and 100 and a pass percentage between 0 and 1).
- Confirmed that the Grade and Result fields are available for the pass/fail and grade distribution visuals.



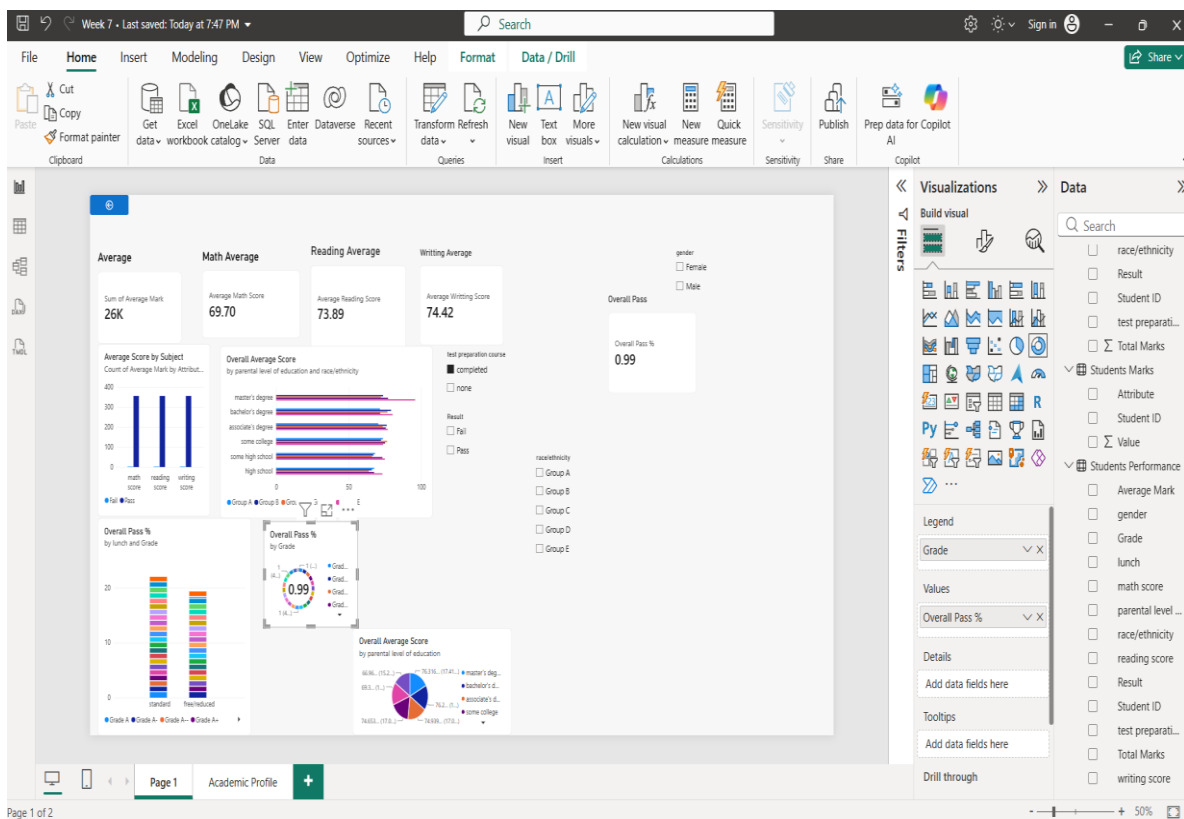
3 Dashboard Questions

Dashboard question	Visual(s) that answer it
What is the overall academic performance?	KPI cards: Average, Math, Reading, Writing Average and Overall Pass
Which subject has the highest or lowest average?	Average Score by Subject column chart; subject KPI cards
Does test preparation relate to performance?	Test preparation course slicer, applied to all KPI cards and charts
How does parental education relate to scores?	Overall Average Score by parental level of education (clustered bar and pie)
How does lunch type relate to performance?	Overall Pass % by lunch and Grade (stacked column chart); lunch slicer
Which groups have higher or lower pass rates?	Overall Pass % by Grade (donut); slicers for gender, race/ethnicity and result

4. KPI Cards

Five KPI cards were placed along the top of the dashboard so that the headline numbers are the first thing a reader sees. Each card is driven by a measure and responds to every slicer on the page.

Card	Field / measure	Purpose
Average	Sum of Average Mark	Headline score indicator
Math Average	Average Math Score	Mean of the math scores
Reading Average	Average Reading Score	Mean of the reading scores
Writing Average	Average Writing Score	Mean of the writing scores
Overall Pass	Overall Pass %	Share of students with a Pass result



5. Subject Performance and Parental Education Analysis

5.1 Subject Performance

A clustered column chart titled Average Score by Subject compares the three subjects. The Attribute field (math score, reading score, writing score) is used on the axis and the value is the average mark, with titles and data labels added for readability. Together with the subject KPI cards, this chart answers which subject is strongest and which is weakest.

On the unfiltered dashboard the subject averages are Math 66.09, Reading 69.17 and Writing 68.05. Reading is the highest and Math the lowest of the three.

5.2 Parental Education

A clustered bar chart titled Overall Average Score by parental level of education and race/ethnicity was created with the following configuration.

Setting	Value
Visual type	Clustered bar chart
Axis	parental level of education (master's degree, bachelor's degree, associate's degree, some college, some high school, high school)
Values	Overall Average Score
Legend	race/ethnicity (Group A to Group E)

A companion pie chart, Overall Average Score by parental level of education, gives a quick share view. Hovering over a bar shows a tooltip with the exact value, for example some high school in Group C has an Overall Average Score of 66.14 . The pie chart labels show the group averages ranging from about 63.1 to 73.6 across the education levels.

Master's degree and bachelor's degree groups sit towards the upper end of the range, while high school and some high school groups sit towards the lower end. This is an association observed in the data and is not evidence that parental education causes the difference.

6. Test Preparation, Lunch Type and Grade Distribution

6.1 Test Preparation

Test preparation is explored with a slicer on the field test preparation course with the values completed and none. Selecting a value updates the KPI cards and charts together, so the averages and pass percentage for each group can be compared directly. This slicer is selected as completed , where the subject averages and pass percentage are higher than the unfiltered values shown

Measure	Completed	All students	Difference
Math average	69.70	66.09	+3.61
Reading average	73.89	69.17	+4.72
Writing average	74.42	68.05	+6.37
Overall Pass %	0.99	0.97	+0.02

6.2 Lunch Type (Socioeconomic Factor)

A stacked column chart titled Overall Pass % by lunch and Grade compares the two lunch types, standard and free/reduced, with each segment showing a grade. A lunch slicer lets the reader see the KPI cards for each lunch type separately, covering Average Score and Pass %.

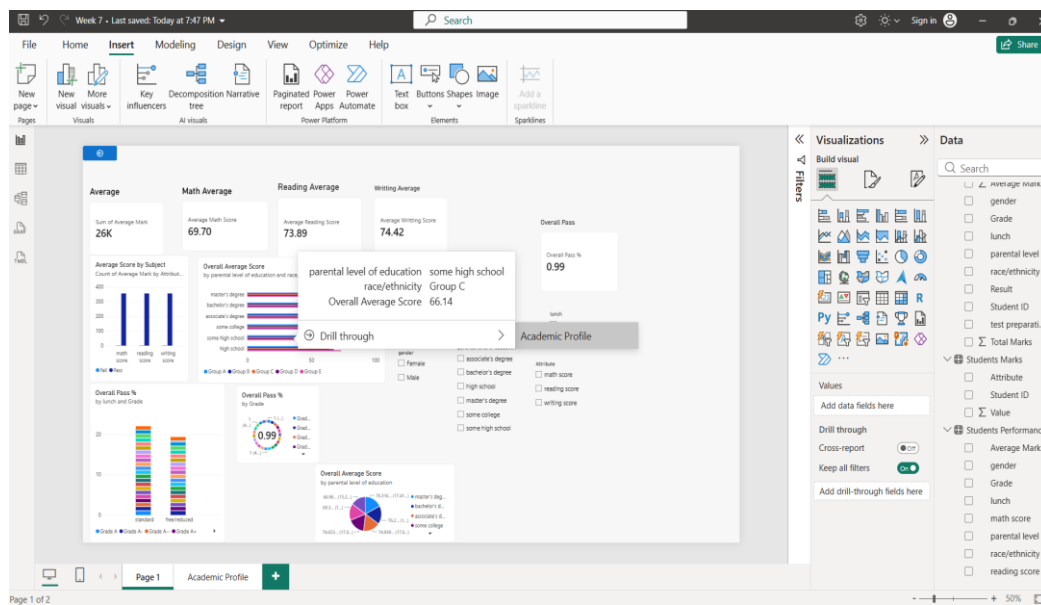
6.3 Grade Distribution

A donut chart titled Overall Pass % by Grade displays how results are spread across the grade categories, with the overall pass value (0.97 in Figure 5) in the centre. The grade field in this dataset is detailed (for example Grade A++, A, A-, B+, down to F), so the legend lists many categories rather than four. If a simplified A, B, C and F view is required, a grouped grade column can be added to the model.

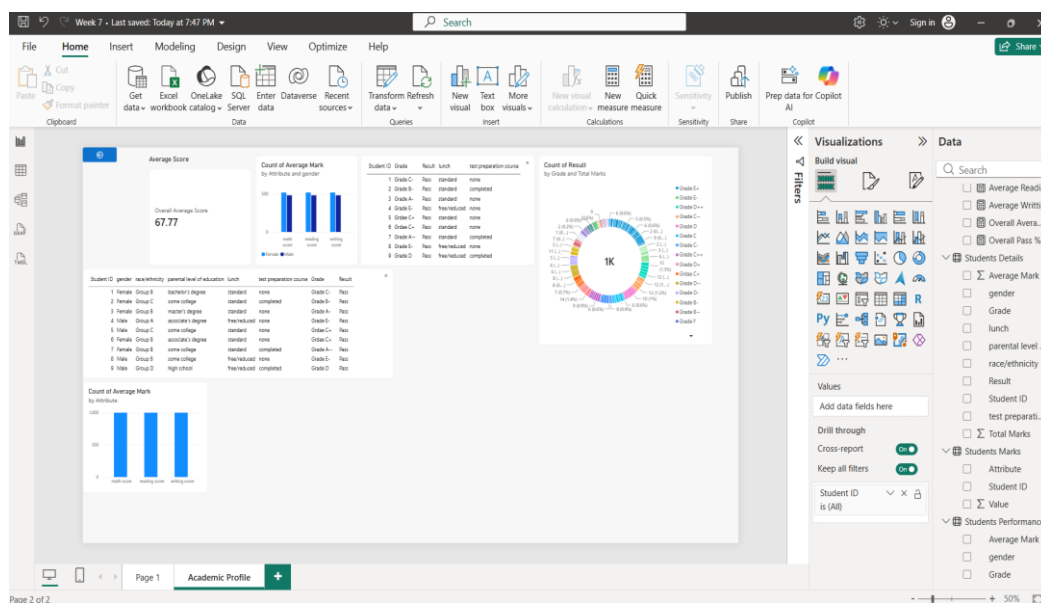
7. Drill-Through Page: Academic Profile

A second page named Academic Profile was added so that a reader can move from the summary dashboard to the detail behind a selected value.

This Figure shows the drill-through in use: right-clicking a bar on the Overall Average Score chart shows the tooltip (some high school, Group C, 66.14) and a Drill through option leading to Academic Profile.



This Figure shows the Academic Profile page. It contains a card for the Overall Average Score (67.77), a column chart of Count of Average Mark by Attribute and gender, a student table (Student ID, Grade, Result, lunch, test preparation course), a detailed table of student attributes, a donut chart of Count of Result by Grade and Total Marks, and a count by Attribute chart.

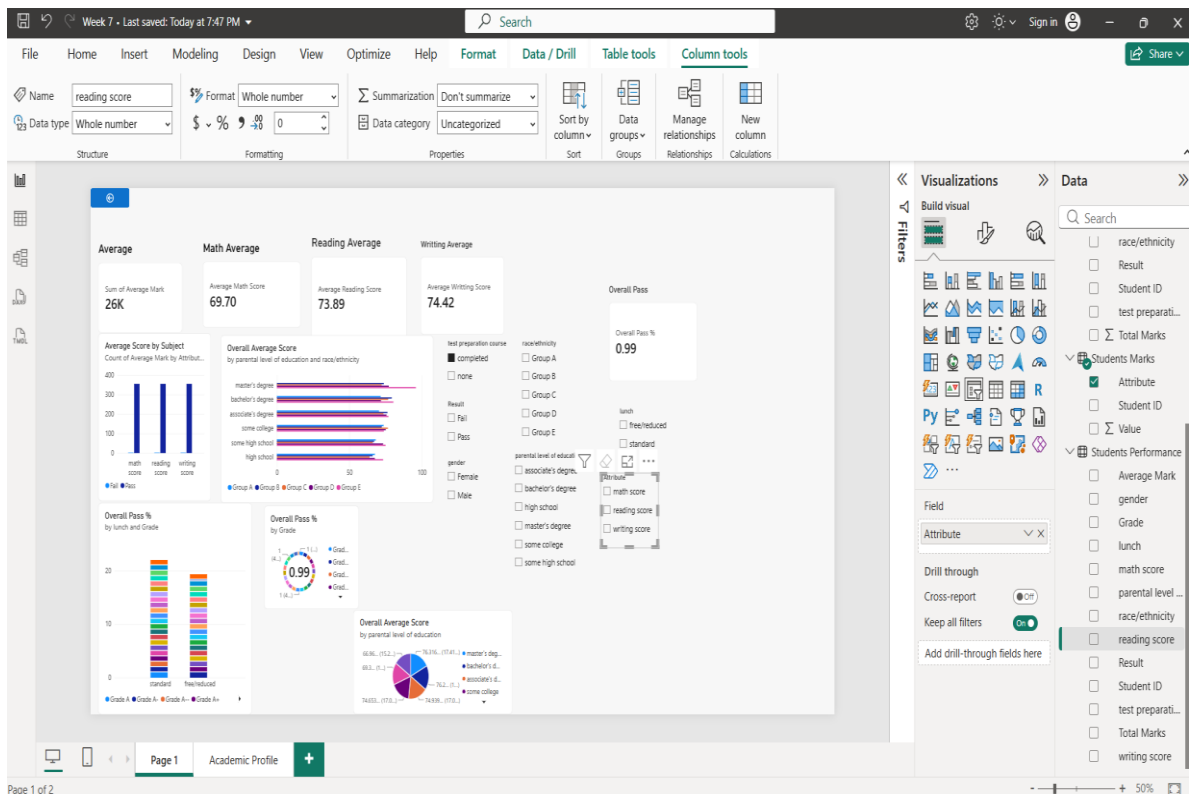


8. Interactive Slicers and Subject Selector

Slicers make the dashboard interactive. Each slicer was formatted as a compact list with a clear title and placed beside the visuals it affects.

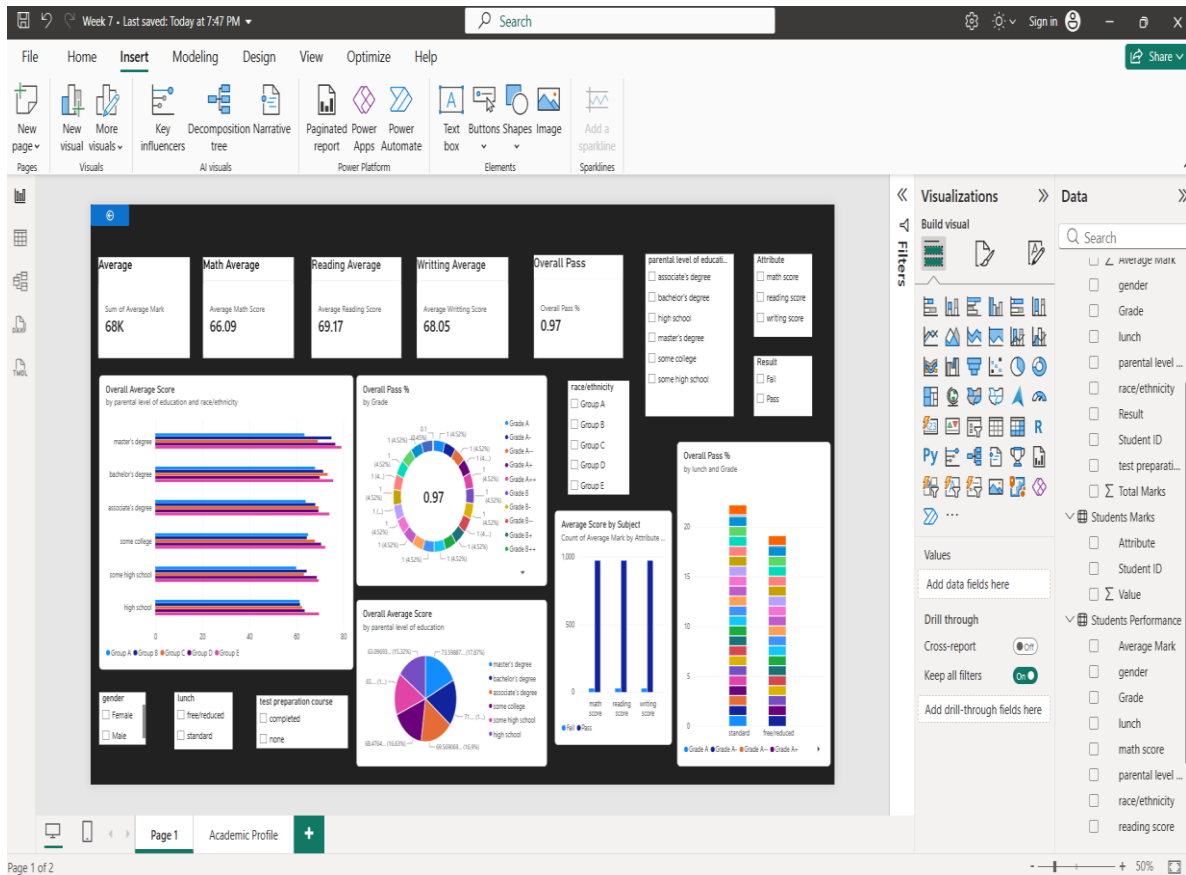
Slicer	Use
gender	Compare Female and Male performance
race/ethnicity	Compare Group A to Group E
parental level of education	Filter by associate's, bachelor's, high school, master's, some college, some high school
lunch	Compare standard and free/reduced
test preparation course	Compare completed and none
Result	Show Fail or Pass students only
Attribute (subject)	Select math score, reading score or writing score

This Figure shows the subject slicer being built. The slicer uses the Attribute field from the Students Marks table, and the Column tools ribbon is open for the reading score column, showing Data type as Whole number and Summarization as Don't summarize. The Attribute slicer lists math score, reading score and writing score, allowing the entire page to be filtered to one subject.



9. Dashboard Formatting and Theme

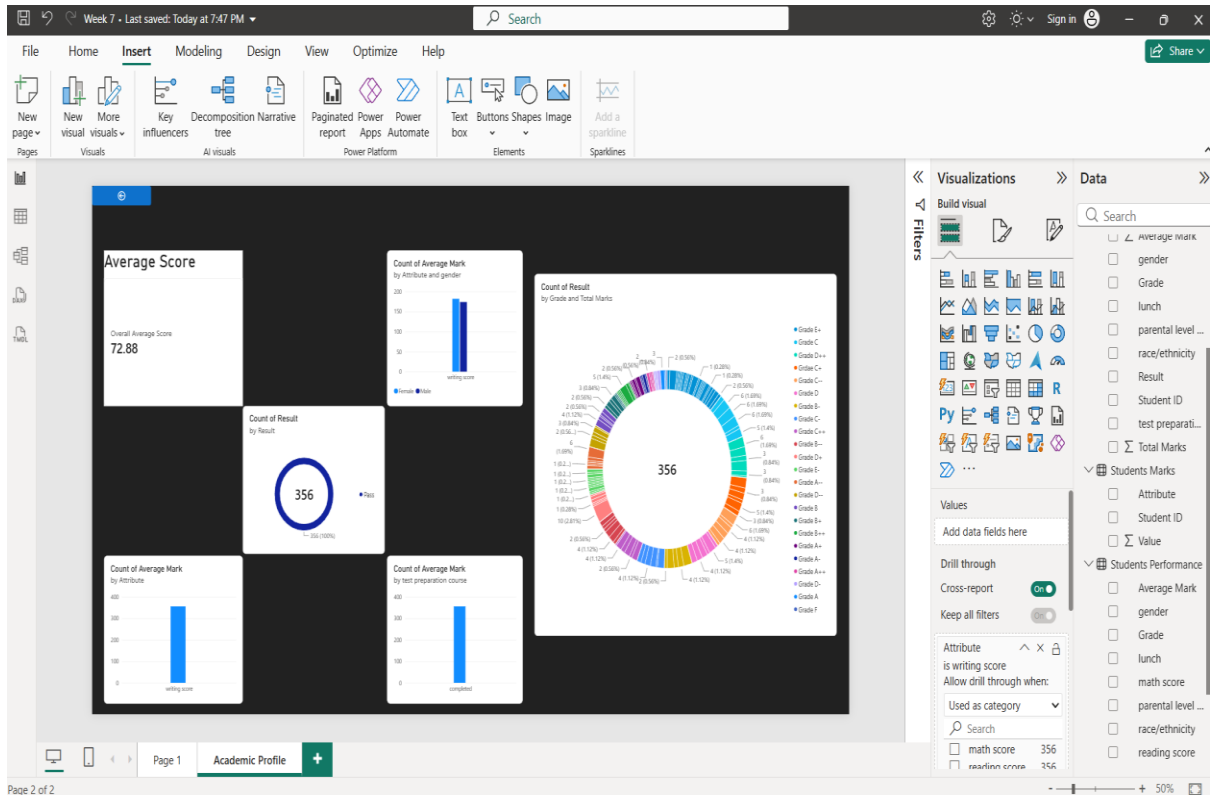
Formatting was applied so that the dashboard is readable at a glance. A dark page background was used so that the white cards and charts stand out, and the same typeface and card style were repeated across all visuals.



10. Academic Profile Detail and Back Button

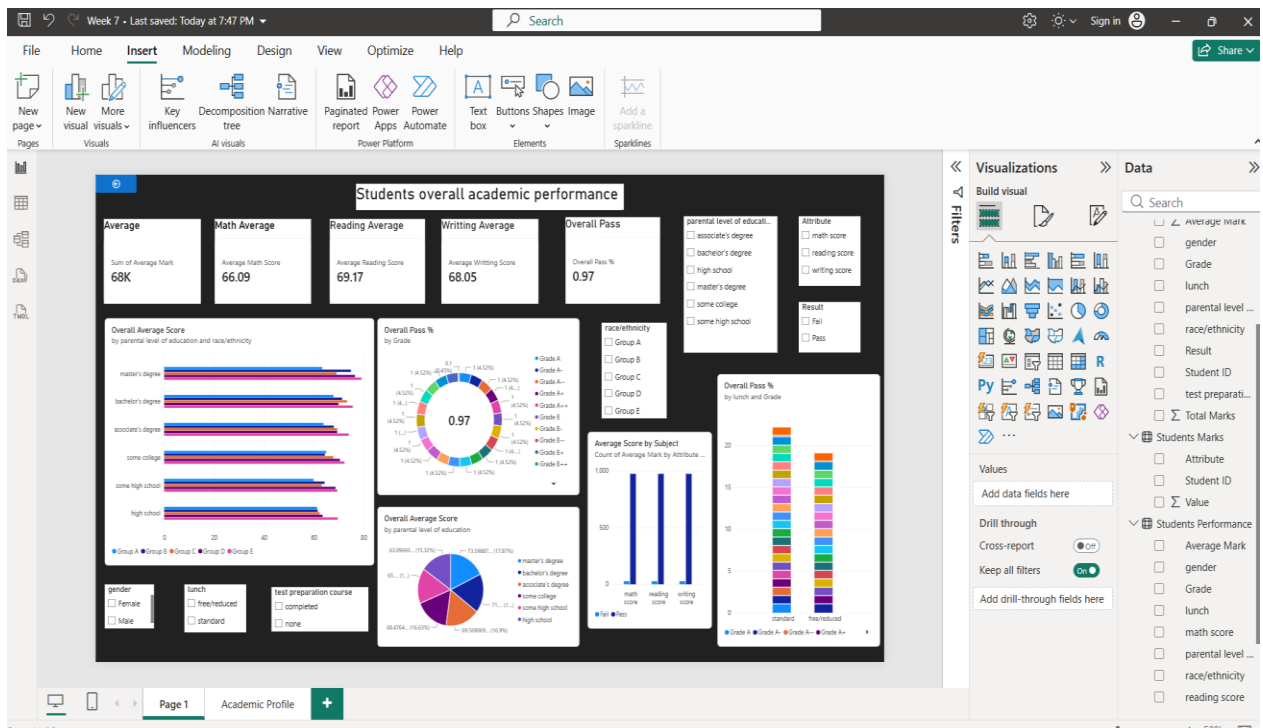
The Academic Profile page is formatted to match the main dashboard. A blue Back button in the top-left corner of the page returns the reader to the main dashboard, which satisfies the requirement to return to the dashboard after drilling through.

This Figure shows the Academic Profile page with the drill-through filter applied. The Filters pane shows the Attribute field filtered to writing score, and the drill-through settings allow drill through when the value is used as a category. In this view the profile shows an Overall Average Score of 72.88, and the donut chart shows a Count of Result of 356, all with a Pass result (100%).



11.Final Dashboard

This Figure shows the finished main dashboard with the title Students overall academic performance added at the top. It combines the KPI cards, comparison charts, slicers and drill-through navigation described in this report.



12. Conclusion

The dashboard answers the questions defined at the start of the activity. The KPI cards give a quick view of overall performance, the subject and parental education charts show where scores differ, and the slicers allow groups to be compared. The Academic Profile page provides a drill-through route to detailed student information, with a Back button to return to the summary.

The main findings are that reading is the highest-scoring subject, math is the lowest, test preparation completion is associated with higher scores and pass rates, and scores vary by parental education. These are associations in one dataset and should not be treated as proof of causes.